

# ECEN 662:

## Estimation & Detection Theory

### Lecture 12: Bayes Estimators and Average-Risk Optimality

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  - Moving from fixed parameters to random variables
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- **Act I: The Philosophy & The Framework**
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  - Defining the *Bayesian Risk*
- **Act II: Deriving Estimators & The Bias Tradeoff**
  - The pointwise optimization trick (Posterior Risk)
  - Conjugate priors and the reality of *shrinkage*
  - Why Bayes estimators are inherently *biased*

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  - Conjugate priors and the reality of *shrinkage*
  - Why Bayes estimators are inherently *biased*
- **Act III: The Bridge to Classical Statistics**
  - *Improper priors* and total ignorance
  - Recovering the frequentist MLE and UMREE (Pitman)

# Setting the Stage: The Bayesian Framework

- **The Goal: Average-Risk Optimality**

In this lecture, we are moving beyond standard estimators to find *average-risk optimality* using the *Bayesian* decision framework.

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- **The Bayesian Shift (The “New World”)**

- We assign a probability measure over the parameter space  $\Omega$ , called the *prior distribution*,  $\pi(\theta)$ .
- We now treat the parameter as a *random variable*,  $\Theta \sim \pi(\theta)$ .

- **The Conditional Risk**

- Under this new generative model, the data is drawn conditionally:  
 $X \sim f(x|\theta) = f_\theta(x)$ .
- Consequently, our familiar frequentist risk is now formally defined as a *conditional expectation*:

$$R_\delta(\theta) = \mathbb{E}_{X|\Theta}[L(\Theta, \delta(X)) \mid \Theta = \theta]$$

# Average Risk Optimality & The Bayes Estimator

- **Averaging Out the Unknown**

To solve the problem of the frequentist risk  $R_\delta(\theta)$  depending on the unknown true parameter, we simply average it out over our prior distribution  $\pi(\theta)$ .

# Average Risk Optimality & The Bayes Estimator

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- **Deriving the Bayes Risk ( $R_{\delta,\pi}$ )**

By applying the Law of Total Expectation, we collapse the conditional frequentist risk into a single, globally defined scalar called the *Bayesian risk*:

$$\begin{aligned} R_{\delta,\pi} &:= \underbrace{\mathbb{E}_\Theta[R_\delta(\Theta)]}_{\text{Average of Frequentist Risk}} \\ &= \underbrace{\mathbb{E}_\Theta[\mathbb{E}_{X|\Theta}[L(\Theta, \delta(X)) \mid \Theta]]}_{\text{Law of Total Expectation}} \\ &= \underbrace{\mathbb{E}_{\Theta,X}[L(\Theta, \delta(X))]}_{\text{Joint Expectation}} \end{aligned}$$

- **The Bayes Estimator ( $\delta_\pi$ )**

Because the Bayesian risk  $R_{\delta,\pi}$  is a single, deterministic number (no longer a function of  $\theta$ ), we can globally minimize it.

- Any estimator  $\delta(X)$  that achieves this absolute minimum is called a *Bayes estimator*, denoted as  $\delta_\pi$ .

# Why Bayes? The Guarantee of Admissibility

- **The Flaw in Classical Methods**

Previous frameworks like the Maximum Likelihood Estimate (MLE) and Uniformly Minimum Variance Unbiased Estimator (UMVUE) frequently produce *inadmissible* estimators.

- Reminder: An estimator is inadmissible if another estimator exists that has lower risk for some  $\theta$ , and never has worse risk for any  $\theta$ .

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- **The Bayesian Guarantee**

By design, Bayes estimators are highly robust against this flaw. We will prove shortly that if a Bayes estimator is *unique*, it is mathematically guaranteed to be *admissible*.

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- **The Complete Class Theorem**

This relationship is surprisingly profound. Under very weak mathematical conditions on the decision problem:

- *Every* admissible estimator is either a Bayes estimator or a limit of Bayes estimators.

- **The Strategic Takeaway**

If your engineering goal is to deploy an optimal, admissible estimator, you can safely restrict your search exclusively to the Bayesian framework.

- Roadmap: As we will see later, finding a highly conservative *minimax* estimator almost always requires finding a Bayes estimator first!

# Theorem: Unique Bayes Estimators are Admissible

- **The Premise**

Suppose we find an estimator  $\delta_\pi$  that minimizes the Bayes risk for a given prior  $\pi$ , and we can prove it is the *unique* minimum.

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- **The Proof (By Contradiction)**

Assume the opposite: suppose our unique Bayes estimator  $\delta_\pi$  is actually *inadmissible*.

- Step 1 (The Dominating Estimator): By definition of inadmissibility, there must exist some other estimator  $\delta'$  that dominates it. For all possible parameters  $\theta \in \Omega$ :

$$R_{\delta'}(\theta) \leq R_{\delta_\pi}(\theta)$$

- Step 2 (Averaging over the Prior): If we take the expected value of both sides over our prior distribution  $\Theta \sim \pi(\theta)$ , the inequality must hold:

$$R_{\delta', \pi} \leq R_{\delta_\pi, \pi}$$

- Step 3 (The Contradiction): Because  $\delta_\pi$  is a Bayes estimator, no estimator can have a strictly lower Bayes risk. Thus, the risks must be equal ( $R_{\delta', \pi} = R_{\delta_\pi, \pi}$ ). This means  $\delta'$  is also a Bayes estimator, which completely violates our premise that  $\delta_\pi$  was unique.

- **A Critical Caveat**

Do not assume the converse! A Bayes estimator can still be admissible even if it is *not* unique.

- Example: If the parameter space  $\Omega$  is strictly *discrete*, any Bayes estimator is guaranteed to be admissible, as long as the prior assigns a non-zero probability to every possible state.

# How to Compute the Bayes Estimator

- **Reversing the Condition: The Posterior Distribution**

Instead of averaging over the data first, let's look at the problem after the data  $X = x$  has been observed. We use Bayes' Rule to find the posterior distribution:

$$\pi(\theta|x) = \frac{\pi(\theta)f(x|\theta)}{\int_{\Omega} \pi(\theta)f(x|\theta)d\theta}$$

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- **The Posterior Risk**

We can now define the expected loss strictly with respect to this new posterior distribution. This is the *posterior risk*:

$$R_{\delta,\pi}(x) := \mathbb{E}_{\Theta|X}[L(\Theta, \delta(x)) \mid X = x]$$

- **The Pointwise Optimization Trick**

By the Law of Total Expectation, the total Bayesian risk  $R_{\delta,\pi}$  is simply the average of the posterior risk over the marginal distribution of the data (the *evidence*,  $m(x)$ ):

$$R_{\delta,\pi} = \mathbb{E}_X[R_{\delta,\pi}(X)]$$

Because the evidence  $m(x)$  is always non-negative, to minimize the massive total integral, we just need to minimize the posterior risk individually for each *specific data point*  $x$ .

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- **The Practical Computation**

To find the Bayes estimator  $\delta_\pi(x)$ , you do not need to solve a functional optimization problem. For any observed  $x$ , simply pick the decision  $d$  that minimizes the posterior risk:

$$\delta_\pi(x) \in \arg \min_d \mathbb{E}_{\Theta|X}[L(\Theta, d) \mid X = x]$$

- **The Fundamental Link**

If  $T(X)$  is a sufficient statistic for the parameter  $\theta$ , then any Bayes estimator  $\delta_\pi(x)$  will automatically depend on the data  $x$  only through  $T(x)$ .

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# Bayes Estimators & Sufficient Statistics

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- **The Proof via Neyman-Fisher**

Let's look at the posterior risk we derived on the last slide. By expanding the posterior distribution and applying the *Neyman-Fisher Factorization Theorem* ( $f_\theta(x) = h(x)g_\theta(T(x))$ ), we get:

$$\begin{aligned}\mathbb{E}_\pi[L(\Theta, d) \mid X = x] &= \int_{\Omega} L(\theta, d) \pi(\theta|x) d\theta \\ &= \frac{1}{m(x)} \int_{\Omega} L(\theta, d) \pi(\theta) f_\theta(x) d\theta \\ &= \frac{h(x)}{m(x)} \int_{\Omega} L(\theta, d) \pi(\theta) g_\theta(T(x)) d\theta\end{aligned}$$

- **The Optimization Constraint**

Notice that the scaling factor  $\frac{h(x)}{m(x)}$  does not depend on our decision variable  $d$ . Therefore, to minimize the posterior risk with respect to  $d$ , we only need to minimize the core integral:

$$\arg \min_d \int_{\Omega} L(\theta, d) \pi(\theta) g_{\theta}(T(x)) d\theta$$

Because the data  $x$  only appears inside the sufficient statistic  $T(x)$  within this integral, the optimal decision  $d$  *must strictly be a function of  $T(x)$* .

# Squared-error loss

- **The Posterior Risk is Quadratic**

Under squared-error loss, the risk with respect to our estimate  $d$  forms a *parabola*:

$$\mathbb{E}_{\Theta|X}[(\Theta - d)^2 | X = x] = d^2 - 2\mathbb{E}_{\Theta|X}[\Theta | X = x]d + \mathbb{E}_{\Theta|X}[\Theta^2 | X = x]$$

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- **The Unique Bayes Estimator**

Because the risk is strictly quadratic, the minimum is *unique*. This gives us the *posterior mean* (often called the *MMSE estimator* in engineering):

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- **The Computational Bottleneck**

Calculating the posterior mean requires the full posterior distribution  $\pi(\theta|x)$ . This is often *analytically impossible* due to the *evidence integral*:

$$m(x) = \int_{\Omega} \pi(\theta) f(x|\theta) d\theta$$

- **The Workaround: Conjugate Priors**

- Problem: High-dimensional integrals are *intractable*.
- Solution: Choosing a *conjugate prior*  $\pi(\theta)$  allows us to bypass the integral and find the posterior mean in *closed-form*.

# The Impact of the Loss Function

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- **Absolute-Error Loss:**  $L(\theta, d) = |\theta - d|$ 
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  - $\delta_\pi(x) = \text{Median}(\Theta|X = x)$  (*Posterior Median*)

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  - More robust to extreme outliers.
  - $\delta_\pi(x) = \text{Median}(\Theta|X = x)$  (*Posterior Median*)
- **Zero-One (Hit-or-Miss) Loss:**  $L(\theta, d) = \mathbb{I}_{\{\theta \neq d\}}$ 
  - Penalizes all errors equally, regardless of distance.
  - $\delta_\pi(x) = \arg \max_\theta \pi(\theta|x)$  (*Maximum a Posteriori / MAP*)

# Conjugate priors

- **The Core Definition**

A family of prior distributions is *conjugate* to a specific likelihood if the resulting posterior belongs to the same family.

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- The Useful Case: Restricting the prior to a specific parameterized family.

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- **Why it matters**

Bayesian updating reduces to simply updating the *parameters* of the distribution, completely *avoiding* the intractable evidence integral.

- **Key Examples**

We will examine two highly practical conjugacies:

- *Multinomial-Dirichlet* (common in machine learning and text processing)
- *Gaussian-Gaussian* (the foundation of signal processing and control)

# Multinomial-Dirichlet Conjugacy (The Setup)

- **The Data Model: Categorical Observations**

Suppose we have  $n$  independent observations  $X = (X_1, \dots, X_n)$  falling into  $K$  known categories:

$$X_i \stackrel{iid}{\sim} \text{Cat}(K, \theta_1, \dots, \theta_K),$$

where our goal is to estimate the unknown probability vector  $\theta = (\theta_1, \dots, \theta_K)$ .

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- **The Multinomial Likelihood**

Instead of the raw sequence, the likelihood is completely defined by the empirical counts  $n_k = \sum_{i=1}^n \mathbf{1}_{\{x_i=k\}}$  for each category:

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- **The Frequentist Baseline (MLE)**

For comparison, recall that the MLE is simply the empirical frequency:

$$\hat{\theta}_k^{\text{MLE}} = \frac{n_k}{n}$$

# The Dirichlet Prior

- **The General Dirichlet Family**

To act as our prior over the probability vector  $\theta$ , we use the *Dirichlet distribution*, parameterized by  $\alpha = (\alpha_1, \dots, \alpha_K)$  where  $\alpha_k > 0$ :

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- **The Concentration Parameter**

Because all elements are identical, the symmetric Dirichlet is parameterized by a *single* scalar  $\alpha$ , known as the *concentration parameter*.

# The Effect of the Concentration Parameter ( $\alpha$ )

- **The Uniform Case ( $\alpha = 1$ )**
  - The symmetric Dirichlet becomes a *flat* distribution over the  $(K - 1)$ -simplex.
  - All probability vectors are *equally* likely.

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- A nice visualization for  $K = 3$  can be found in this blog

# The Dirichlet Posterior Update

- **The Conjugacy Proof**

To see that the Dirichlet family is *conjugate* to the multinomial likelihood, we multiply them and group the exponents:

$$\begin{aligned}\pi_{\alpha}(\theta|x) &\propto_{\theta} p_{\theta}(x)\pi_{\alpha}(\theta) \\ &\propto_{\theta} \left( \prod_{k=1}^K \theta_k^{n_k} \right) \left( \prod_{k=1}^K \theta_k^{\alpha_k-1} \right) \\ &= \prod_{k=1}^K \theta_k^{n_k+\alpha_k-1}\end{aligned}$$

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- **The Dirichlet Mean**

Recall that we need the *posterior mean* to find our MMSE Bayes estimator. For any  $\Theta \sim \text{Dir}(\alpha)$ , the expected value is:

$$\mathbb{E}\alpha[\Theta_k] = \frac{\alpha_k}{\alpha_0}, \quad k \in [K],$$

where  $\alpha_0 = \sum_{k=1}^K \alpha_k$  is the sum of all concentration parameters.

# The Bayes Estimator: Smoothing & Shrinkage

- **The Posterior Mean**

Using the updated parameters  $(n_k + \alpha_k)$ , the Bayes estimator under squared-error loss is:

$$\mathbb{E}_\alpha[\Theta_k | X = x] = \frac{n_k + \alpha_k}{n + \alpha_0}, \quad \forall k \in [K]$$

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- **Laplacian Smoothing**

When using a *symmetric* Dirichlet prior (e.g.,  $\alpha_k = 1$ ), this estimator is famously known in computer science and engineering as *Laplacian smoothing* (or additive smoothing).

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- **The Convex Combination**

We can algebraically rewrite the posterior mean to reveal its true nature:

$$\mathbb{E}_\alpha[\Theta_k | X = x] = \underbrace{\left( \frac{n}{n + \alpha_0} \right)}_{\text{Weight on Data}} \underbrace{\left( \frac{n_k}{n} \right)}_{\text{MLE}} + \underbrace{\left( \frac{\alpha_0}{n + \alpha_0} \right)}_{\text{Weight on Prior}} \underbrace{\left( \frac{\alpha_k}{\alpha_0} \right)}_{\text{Prior Mean}}$$

- **The Shrinkage Estimator**

- This is a *shrinkage estimator*: it “shrinks” the purely data-driven MLE toward our prior belief.
- *Asymptotic Equivalence*: As  $n \rightarrow \infty$ , the data overwhelms the prior, and the Bayes estimator converges to the MLE.

# Gaussian Conjugacy (Unknown Mean)

- **The Data Model: Gaussian Noise**

Suppose we have  $n$  independent observations  $X = (X_1, \dots, X_n)$  drawn from a Gaussian distribution:

$$X_i \stackrel{iid}{\sim} \mathcal{N}(\mu, \sigma^2),$$

where the variance  $\sigma^2$  is known (e.g., calibrated sensor noise), but the mean  $\mu$  is the *unknown* parameter we want to estimate.

# Gaussian Conjugacy (Unknown Mean)

- **The Data Model: Gaussian Noise**

Suppose we have  $n$  independent observations  $X = (X_1, \dots, X_n)$  drawn from a Gaussian distribution:

$$X_i \stackrel{iid}{\sim} \mathcal{N}(\mu, \sigma^2),$$

where the variance  $\sigma^2$  is known (e.g., calibrated sensor noise), but the mean  $\mu$  is the *unknown* parameter we want to estimate.

- **The Likelihood & Frequentist Baseline**

- The joint likelihood of our data is:

$$f_{\mu}(x) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2\right)$$

- The Baseline (MLE): Recall that maximizing this likelihood yields the standard sample mean:

$$\hat{\mu}_{\text{MLE}} = \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

- **The Gaussian Prior**

To set up our Bayesian framework, we assign a Gaussian prior to the unknown mean  $\mu$ :

$$\pi_{\mu_0, \sigma_0^2}(\mu) = \frac{1}{\sqrt{2\pi\sigma_0^2}} \exp\left(-\frac{(\mu - \mu_0)^2}{2\sigma_0^2}\right)$$

parameterized by our prior mean  $\mu_0$  and prior variance  $\sigma_0^2$ .

# Gaussian Conjugacy (The Posterior Update)

- **Multiplying the Exponentials**

To find the posterior, we multiply the Gaussian prior and the Gaussian likelihood. Because both are exponentials, we simply add their exponents and ignore any constants that do not depend on  $\mu$ :

$$\pi_{\mu_0, \sigma_0^2}(\mu|x) \propto_{\mu} \exp\left(-\frac{1}{2\sigma_0^2}(\mu^2 - 2\mu_0\mu)\right) \exp\left(-\frac{n}{2\sigma^2}(\mu^2 - 2\bar{x}\mu)\right)$$

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- **Grouping and Completing the Square**

By grouping the  $\mu^2$  and  $\mu$  terms, we construct a new quadratic in the exponent:

$$\propto_{\mu} \exp\left(-\underbrace{\left(\frac{1}{2\sigma_0^2} + \frac{n}{2\sigma^2}\right)}_{\text{New inverse-variance}} \mu^2 + \underbrace{\left(\frac{\mu_0}{\sigma_0^2} + \frac{n\bar{x}}{\sigma^2}\right)}_{\text{Shift term}} \mu\right)$$

Completing the square on this expression returns it to the standard Gaussian form of  $\exp(-a(\mu - b)^2)$ .

- **The Posterior Mean Formula**

By separating the terms of our updated mean  $\mu_n$ , we can rewrite the Bayes estimator under squared-error loss:

$$\mu_n = \frac{\mu_0\sigma^2 + n\sigma_0^2\bar{x}}{\sigma^2 + n\sigma_0^2} = \frac{n\sigma_0^2}{\sigma^2 + n\sigma_0^2}\bar{x} + \frac{\sigma^2}{\sigma^2 + n\sigma_0^2}\mu_0$$

# The Bayes Estimator: Gaussian Shrinkage

- **The Exact Gaussian Posterior**

The algebra proves the posterior is a new Gaussian distribution,  $\mathcal{N}(\mu_n, \sigma_n^2)$ , with updated parameters:

- Posterior Mean ( $\mu_n$ ):  $\frac{\mu_0\sigma^2+n\sigma_0^2\bar{x}}{\sigma^2+n\sigma_0^2}$
- Posterior Variance ( $\sigma_n^2$ ):  $\frac{\sigma_0^2\sigma^2}{\sigma^2+n\sigma_0^2}$

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- **The Convex Combination**

Just like the Multinomial-Dirichlet case, this is a *shrinkage estimator*. It is a weighted average of the data ( $\bar{x}$ ) and the prior ( $\mu_0$ ):

$$\mu_n = \underbrace{\left( \frac{n \sigma_0^2}{\sigma^2 + n \sigma_0^2} \right)}_{\text{Weight on MLE}} \bar{x} + \underbrace{\left( \frac{\sigma^2}{\sigma^2 + n \sigma_0^2} \right)}_{\text{Weight on Prior}} \mu_0$$

- **Engineering Intuition (The Limits)**

- Infinite Data ( $n \rightarrow \infty$ ): The weight on the MLE approaches 1. The data overpowers the prior:  $\mu_n \approx \bar{x}$ .
- Terrible Sensor ( $\sigma^2 \rightarrow \infty$ ): The weight on the prior approaches 1. We ignore the noisy data entirely:  $\mu_n \approx \mu_0$ .
- Total Ignorance ( $\sigma_0^2 \rightarrow \infty$ ): The prior is completely flat. We trust the data entirely:  $\mu_n \approx \bar{x}$ .

# The biasedness of Bayes estimators

- **The Reality of Shrinkage**

- In both the Dirichlet and Gaussian examples, the Bayes estimators are inherently *biased* for all finite sample sizes.
- This is a *fundamental* feature, not a bug. They trade a little bias for a massive reduction in variance.

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- **The Reality of Shrinkage**

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- This is a *fundamental* feature, not a bug. They trade a little bias for a massive reduction in variance.

- **The Theorem: Unbiased = Degenerate**

Under squared-error loss, a Bayes estimator  $\delta_\pi(x)$  is only unbiased in the *degenerate* case where the Bayesian risk is exactly *zero* (i.e., perfect estimation, which is impossible in noisy environments).

- **The Proof (By Contradiction)**

Assume the Bayes estimator is unbiased:  $\mathbb{E}[\delta_\pi(X)|\Theta = \theta] = \theta$  for all  $\theta$ . Let's evaluate the cross-term  $\mathbb{E}[\delta_\pi(X)\Theta]$  in two different ways using the Law of Total Expectation:

- Path A (Conditioning on  $\Theta$ ):

$$\mathbb{E}[\delta_\pi(X)\Theta] = \mathbb{E}[\mathbb{E}[\delta_\pi(X)|\Theta]\Theta] = \mathbb{E}[\Theta \cdot \Theta] = \mathbb{E}[\Theta^2]$$

- Path B (Conditioning on  $X$ ):

$$\mathbb{E}[\delta_\pi(X)\Theta] = \mathbb{E}[\delta_\pi(X)\mathbb{E}[\Theta|X]] = \mathbb{E}[\delta_\pi(X)\delta_\pi(X)] = \mathbb{E}[\delta_\pi^2(X)]$$

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- **The Conclusion (Zero Risk)**

Expanding the Bayesian risk and substituting Path A and Path B:

$$\begin{aligned}\mathbb{E}_\pi[(\delta_\pi(X) - \Theta)^2] &= \mathbb{E}_\pi[\delta_\pi^2(X)] - 2\mathbb{E}_\pi[\delta_\pi(X)\Theta] + \mathbb{E}_\pi[\Theta^2] \\ &= \underbrace{(\mathbb{E}_\pi[\delta_\pi^2(X)] - \mathbb{E}_\pi[\delta_\pi(X)\Theta])}_{=0 \text{ (from Path B)}} + \underbrace{(\mathbb{E}_\pi[\Theta^2] - \mathbb{E}_\pi[\delta_\pi(X)\Theta])}_{=0 \text{ (from Path A)}} = 0\end{aligned}$$

# Is the Sample Mean a Bayes Estimator?

- **The Diagnostic Test**

We can use our previous result as a strict test: Under squared-error loss, if an estimator is *unbiased* but has *non-zero* risk, it cannot be a Bayes estimator.

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- **The Test Case: Gaussian Mean**

Let's test the standard frequentist Maximum Likelihood Estimate (MLE). Can the sample mean  $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$  be the Bayes estimator for some prior  $\pi$  under squared-error loss?

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- **Running the Test**

- Condition 1 (Unbiased):  $\bar{X}$  is an unbiased estimator of  $\mu$ .
- Condition 2 (Non-zero Risk): Its Mean Squared Error (MSE) is strictly positive for any finite sample size  $n$ :

$$\mathbb{E}_{\mu}[(\bar{X} - \mu)^2] = \frac{\sigma^2}{n} > 0, \quad \forall \mu$$

Because the MSE is positive for all  $\mu$ , the overall Bayesian risk will be non-zero for any prior  $\pi$ .

- **The Conclusion**

Because it is unbiased but has a non-zero risk,  $\bar{X}$  *cannot* be the true Bayes estimator for any proper prior  $\pi$  under squared-error loss.

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- **A Critical Caveat**

This mutual exclusivity between “unbiased” and “Bayes” is a special property of the *squared-error loss*. Under other loss functions (like absolute-error), it is mathematically possible to have an unbiased Bayes estimator with a non-zero risk.

# Generalized Bayes estimators

- **The Loophole: Improper Priors**

While  $\bar{X}$  cannot be a standard Bayes estimator, it can be a *generalized* Bayes estimator if we allow the prior distribution to be *improper* (i.e., it does not integrate to 1):

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Surprisingly, multiplying an improper prior by a highly concentrated likelihood can still yield a perfectly valid, *proper posterior*.

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- **The Proof: The Flat Prior**

Let's assign a completely flat, improper prior to our Gaussian mean:  $\pi(\mu) = 1$  for all  $\mu \in \mathbb{R}$ . The posterior update becomes:

$$\begin{aligned}\pi(\mu|x) &\propto_{\mu} \pi(\mu) f_{\mu}(x) \\ &\propto_{\mu} 1 \cdot \exp\left(-\frac{n}{2\sigma^2}(\mu^2 - 2\mu\bar{x})\right) \\ &\propto_{\mu} \exp\left(-\frac{1}{2(\sigma^2/n)}(\mu - \bar{x})^2\right)\end{aligned}$$

- **The Grand Conclusion**

The resulting posterior is exactly  $\mathcal{N}(\bar{x}, \sigma^2/n)$ .

- Under squared-error loss, we take the posterior mean.
- Therefore, the sample mean  $\bar{X}$  is exactly the *generalized* Bayes estimator under a *flat* improper prior.

# Generalized Bayes & The Pitman Estimator

- **The Location Family Model**

Let's generalize beyond the Gaussian. Consider  $n$  independent observations drawn from any shifted distribution:

$$X_i \stackrel{iid}{\sim} g(x_i - \theta)$$

Here,  $g$  is a known base probability density, and  $\theta \in \mathbb{R}$  is the unknown *location parameter* we want to estimate.

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Here,  $g$  is a known base probability density, and  $\theta \in \mathbb{R}$  is the unknown *location parameter* we want to estimate.

- **The General Posterior Mean**

Given an arbitrary prior  $\pi(\theta)$ , the posterior is proportional to the prior times the likelihood:

$$\pi(\theta|x) \propto_{\theta} \pi(\theta) \prod_{i=1}^n g(x_i - \theta)$$

Under squared-error loss, our Bayes estimator is the posterior mean:

$$\mathbb{E}_{\pi}[\Theta|X = x] = \frac{\int_{-\infty}^{\infty} \theta \pi(\theta) \prod_{i=1}^n g(x_i - \theta) d\theta}{\int_{-\infty}^{\infty} \pi(\theta) \prod_{i=1}^n g(x_i - \theta) d\theta}$$

- **The Flat Prior & The Pitman Estimator**

If we model total ignorance by choosing the *improper prior*  $\pi(\theta) = 1$ , the  $\pi(\theta)$  terms vanish from the integrals.

- The resulting formula is exactly the *Pitman estimator of location* as we discussed previously.
- The Grand Connection: The Pitman estimator is famously known as the Uniformly Minimum Risk Equivariant Estimator (UMREE) under squared-error loss.

# The Catch: Limits of Generalized Bayes

- **The Warning**

While generalized Bayes estimators elegantly bridge the gap to classical frequentist methods (like the MLE and Pitman estimator), they do *not* inherit the strict theoretical safety nets of proper Bayes estimators.

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- **Loss of Guaranteed Admissibility**

- Proper Bayes: If a prior is proper and strictly positive, the resulting Bayes estimator is mathematically guaranteed to be *admissible* (no other estimator can strictly beat it for all  $\theta$ ).
- Generalized Bayes: This guarantee vanishes. Because the prior is improper, the estimator can be *inadmissible*.

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- **The Risk Problem**

Because an improper prior integrates to infinity ( $\int_{\Omega} \pi(\theta) d\theta = \infty$ ), the overall integrated Bayesian risk is technically infinite. We cannot rely on standard risk-minimization theorems to prove optimality.

- **The Grand Summary: Which do we choose?**

| Feature             | Proper Bayes          | Generalized Bayes      |
|---------------------|-----------------------|------------------------|
| Prior $\pi(\theta)$ | Integrates to 1       | Integrates to $\infty$ |
| Bias                | Inherently Biased     | Often Unbiased         |
| Admissibility       | Guaranteed Admissible | Can be Inadmissible    |
| Equivalence         | MAP / MMSE Filtering  | MLE / UMREE            |